

Homework06

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Q6. How would you generalize the original code above to work with any set of input protein structures?

```
#' Title: Protein Analysis Plotter. Visualize B-factors from PDB Structures
#
# This function reads a protein structure from the Protein Data Bank (PDB),
# selects atoms either by biological criteria (chain and atom type) or by
# precomputed indices, and plots atomic B-factors with
# optional secondary structure annotation.
#
#' @param pdb_id Character string specifying the PDB accession code.
#'   **Required.**
#' @param chain Character string specifying the chain identifier to select.
#'   **Optional.** Used only if inds is not provided.
#' @param elty Character string specifying the atom or
#' element type (e.g., "CA").
#'   **Optional.** Used only if inds is not provided.
#' @param inds Optional precomputed atom indices obtained from
#' atom.select(). **Optional.** If provided, overrides
#' chain and elty.
#' @param sse Logical indicating whether secondary structure information
#' should be retained during trimming. **Optional.**
#' @param typ Character string specifying the plot type passed to
#' plotb3(). Default set to "l". **Optional.**
#' @param ylab Character string specifying the y-axis label.
#' Default set to "B-factor". **Optional.**
#' @param main Optional character string specifying the plot title.
#'   **Optional.**
#' @param ... Additional graphical parameters passed to plotb3().
#'   **Optional.**
#
# @details
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#' -The only required argument is pdb_id. If pdb_id is a vector,
#' a single inds is typically only valid for one structure.
#' -Atom selection must be specified either
#' by providing inds OR by providing both chain and elety.
#' -If inds is provided, it takes precedence over chain and elety.
#' -If chain or elety is supplied as a vector, only the first element is used.
#' -This ensures a single, consistent selection is applied
#' across all PDB structures processed by the function.
#' @returns
#' Returns the plot of B-factors for the selected atoms
#' (based on chain and atom type or index) in the
#' specified PDB structure.
#' Invisibly returns a list containing the trimmed
#' PDB structure and b-factors for each provided pdb_id.
#' @export
#'
#' @examples
#' protein_analysis_plotter(c("4AKE", "1AKE", "1E4Y"), "A", "CA", sse = FALSE)

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protein_analysis_plotter<- function(
  pdb_id,
  chain = NULL,
  elety = NULL,
  inds = NULL,
  sse = TRUE,
  typ = "l",
  ylab = "B-factor",
  main = NULL,
  ...
){
  # Load required library
  library(bio3d)

  #Required Argument Checks
  if (missing(pdb_id) || length(pdb_id) < 1 || is.null(pdb_id)) {
    stop("`pdb_id` is required and must be a non-empty character string.")
  }
  pdb_id <- as.character(pdb_id)

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#Validate main
if (!is.null(main)) {
  if (!is.character(main)) {
    stop("`main` must be NULL or a character vector.")
  }
  if (!(length(main) == 1 || length(main) == length(pdb_id))) {
    stop("`main` must have length 1 or the same length as `pdb_id`.")
  }
}

#Enforce inds or (chain and elety) paths
using_inds = !is.null(inds)
if(using_inds){
  if (!is.null(chain) || !is.null(elety)) {
    stop("Provide either `inds` OR (`chain` and `elety`), not both.")
  }
}else{
  if (is.null(chain) || is.null(elety)) {
    stop("Either `inds` OR both `chain` and `elety` must be provided.")
  }
  #force all plots to use single chain and elety
  chain <- as.character(chain)[1]
  elety <- as.character(elety)[1]

  #final check
  if (!nzchar(chain) || !nzchar(elety)) {
    stop("`chain` and `elety` must be non-empty character strings.")
  }
}

#store results here
plot_results <- vector("list", length = length(pdb_id))
names(plot_results) <- pdb_id

#loop over each pdb_id, for each do read.pdb, trim.pdb, extract b-factors, plot
for(i in seq_along(pdb_id)){
  pb_id <- pdb_id[i]

  # Read the PDB structure carefully, wrap in tryCatch to handle potential errors
  pdb_structure <- tryCatch(
    read.pdb(pb_id),
    error = function(e) {

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        warning("Skipping PDB '", pb_id, "': failed to read PDB (", e$message, ")")
        return(NULL)
    }
)
if (is.null(pdb_structure)) next

# Select atoms based on provided criteria
# Trim carefully, use tryCatch to handle potential errors
selected_trim <- tryCatch(
  {
    if (using_inds) { #inds methods
      trim.pdb(pdb_structure, inds = inds, sse = sse)
    } else { #chain + elety
      trim.pdb(pdb_structure, chain = chain, elety = elety, sse = sse)
    }
  },
  error = function(e) {
    warning("Skipping PDB '", pb_id, "': trimming failed (", e$message, ")")
    return(NULL)
  }
)
if (is.null(selected_trim)) next

# Check if atoms remain
if (is.null(selected_trim$atom) || nrow(selected_trim$atom) == 0) {
  warning("Skipping PDB '", pb_id, "': no atoms remain after trimming")
  next
}

# Extract B-factors carefully
# Check if atoms has a b column
if (!"b" %in% colnames(selected_trim$atom)) {
  warning("Skipping PDB '", pb_id, "': no B-factor data available")
  next
}

b_factors <- selected_trim$atom$b

# Ensure b_factors is numeric
if (!is.numeric(b_factors)) {
  warning("Skipping PDB '", pb_id, "': B-factor data is not numeric")
  next
}

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}

#If allow pbd_ids to be vecotr then allow main to be vector too
plot_main <- if (is.null(main)) {
  NULL
} else if (length(main) == 1) {
  main
} else {
  main[i]
}

#Create specific title for each plot
if (is.null(plot_main)) {
  plot_main <- paste0("B-factors: ", pb_id)

  if(using_inds){
    plot_main <- paste0(plot_main, " (using provided indices)")
  }else{
    plot_main <- paste0(plot_main, " (Chain: ", chain, ", Elety: ", elety, ")")
  }
}

}

# Plot B-factors
plotb3(b_factors, sse = selected_trim, typ = typ, ylab = ylab, main = plot_main, ...)

# Store results, main goal is to print the plot but store these
# for potential further analysis
plot_results[[pb_id]] <- list(
  pdb = selected_trim,
  b_factors = b_factors
)
}
invisible(plot_results)
}

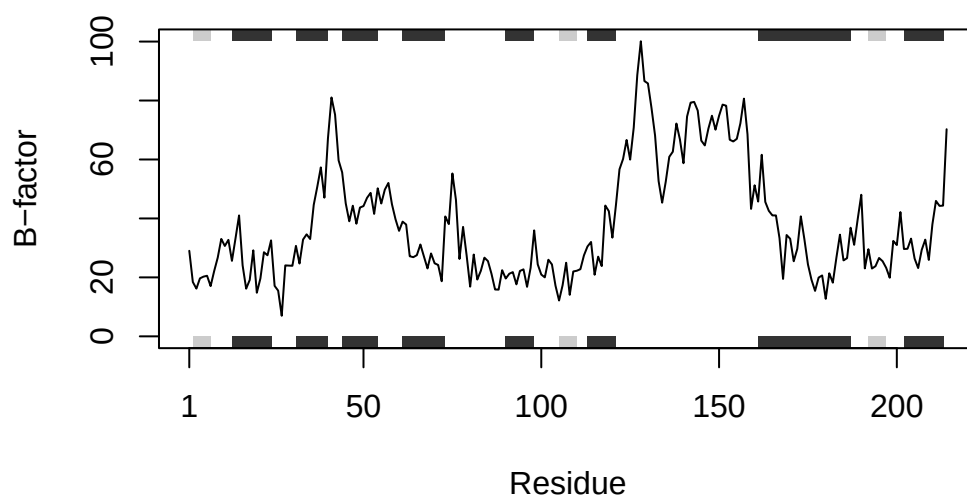
protein_analysis_plotter(c("4AKE", "1AKE", "1E4Y"), "A", "CA", sse = TRUE)

```

Warning: package 'bio3d' was built under R version 4.5.2

Note: Accessing on-line PDB file

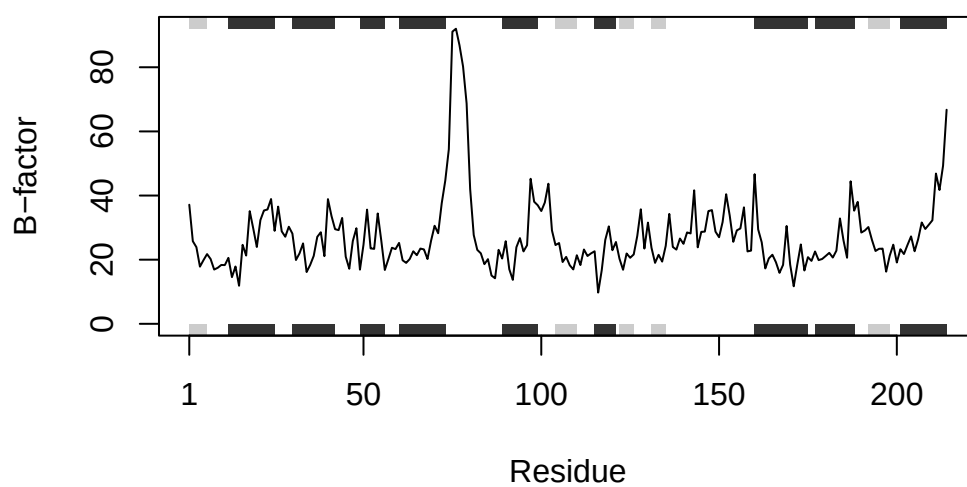
B-factors: 4AKE (Chain: A, Eley: CA)



Note: Accessing on-line PDB file

PDB has ALT records, taking A only, rm.alt=TRUE

B-factors: 1AKE (Chain: A, Eley: CA)



Note: Accessing on-line PDB file

B-factors: 1E4Y (Chain: A, Eley: CA)

